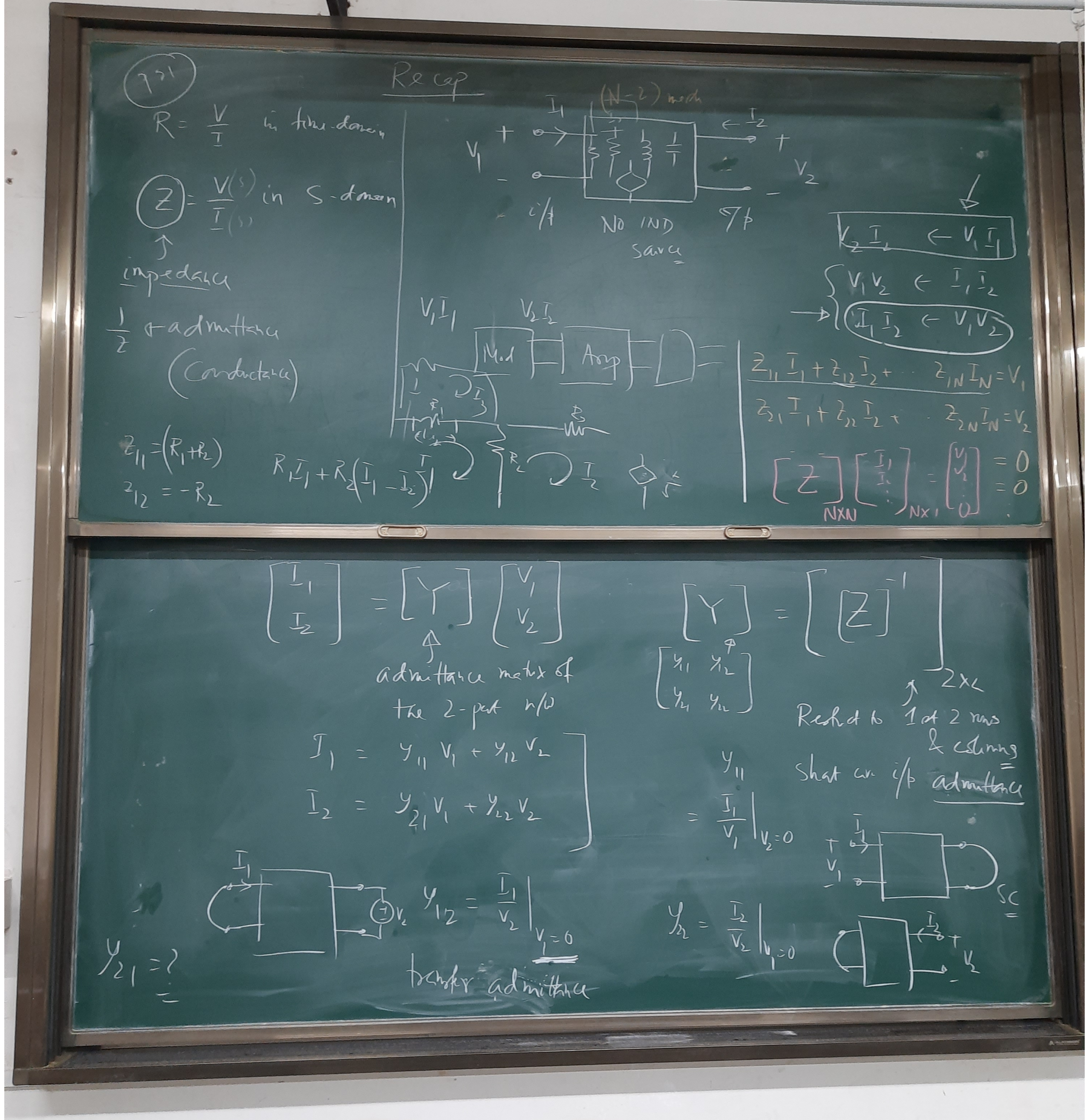


7/1

Recap

$R = \frac{V}{I}$ in time-domain

$\boxed{Z} = \frac{V(s)}{I(s)}$ in S-domain

impedance

$\frac{1}{Z}$ ← admittance

(Conductance)

$Z_{11} = (R_1 + R_2)$

$Z_{12} = -R_2$

$R_1 I_1 + R_2(I_1 - I_2)$

$\boxed{V_2 I_1 \leftarrow V_1 I_1}$

$$\left\{\begin{array}{l} V_1 V_2 \leftarrow I_1 I_2 \\ \boxed{I_1 I_2 \leftarrow V_1 V_2} \end{array}\right.$$

$$Z_{11} I_1 + Z_{12} I_2 + \cdots Z_{1N} I_N = V_1$$

$$Z_{21} I_1 + Z_{22} I_2 + \cdots Z_{2N} I_N = V_2$$

$$[Z]_{N \times N} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \end{bmatrix}_{N \times 1} = \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ 0 \end{bmatrix} \begin{array}{l} = 0 \\ = 0 \end{array}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = [Y] \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

admittance matrix of the 2-port n/w

$$\left.\begin{array}{l} I_1 = y_{11} V_1 + y_{12} V_2 \\ I_2 = y_{21} V_1 + y_{22} V_2 \end{array}\right]$$

$$y_{12} = \left.\frac{I_1}{V_2}\right|_{V_1 = 0}$$

transfer admittance

$y_{21} = ?$

$$[Y] = \Big[[Z]^{-1}\Big]_{2 \times 2}$$

$$\begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix}$$

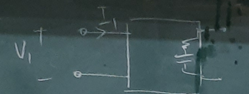
Restricted to 1 of 2 rows & columns

$$y_{11} = \left.\frac{I_1}{V_1}\right|_{V_2 = 0}$$

Short cir. i/p admittance

$$y_{22} = \left.\frac{I_2}{V_2}\right|_{V_1 = 0}$$

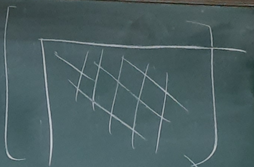
1-port



+1 (N−2)

N-1 mesh

Z_1



$Z_{11} I_1 + Z_{13} I_3 + \dots + Z_{1N} I_N = V_1$

$Z_{21} I_1 + Z_{23} I_3 \dots = 0$

$\vdots$

$Z_{N1} I_1 + Z_{N3} I_3 + \dots = 0$

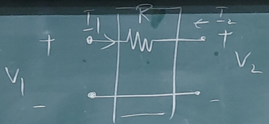
$Z_{in} = \frac{V_1}{I_1}$

$I_1 = \frac{|K|}{|Z_{-1}|} V_1$

P33

$[Z_{-1}] I = \begin{bmatrix} V_1 \\ 0 \\ \vdots \end{bmatrix}$ N-1 X N-1 Cramer's law

P34 Example ①



$[Y] = ?$

Try V/SC combo

(a bit lengthy) set $(y_{11}, y_{12} \dots)$

Faster

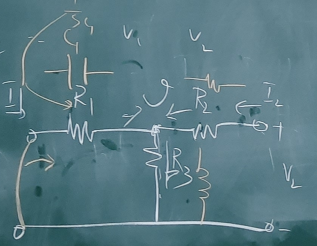
$\begin{cases} I_1 = V_1 - V_2 \\ I_2 \end{cases}$

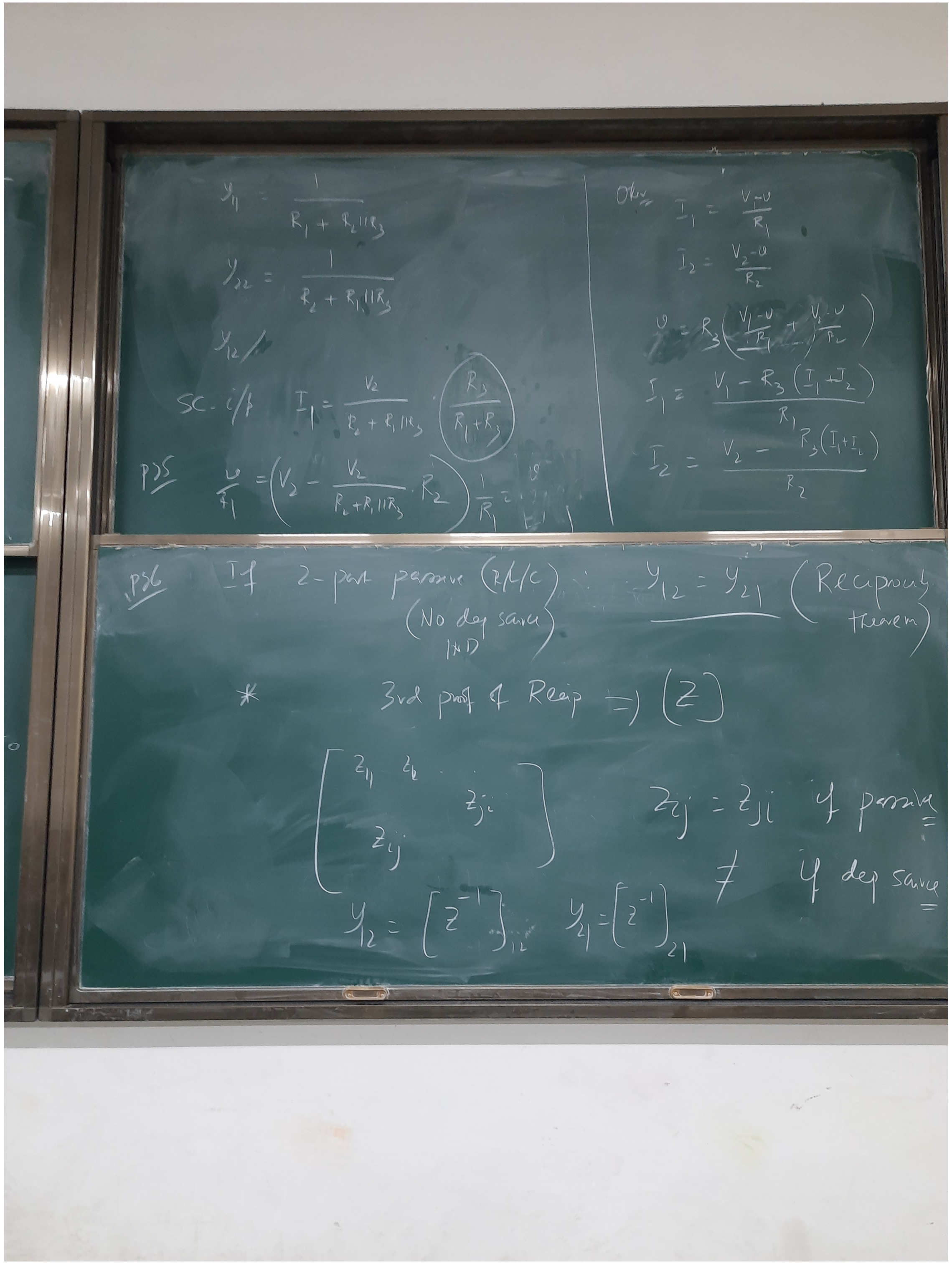
$I_1 = \frac{V_1 - V_2}{R}$

$I_2 = \frac{V_2 - V_1}{R}$

$\begin{bmatrix} \frac{1}{R} & \overset{<0}{-\frac{1}{R}} \\ -\frac{1}{R} & \frac{1}{R} \end{bmatrix}$

②





$$y_{11} = \frac{1}{R_1 + R_2 \| R_3}$$

$$y_{22} = \frac{1}{R_2 + R_1 \| R_3}$$

y_{12} /

SC. i/p $\quad I_1 = \frac{V_2}{R_2 + R_1 \| R_3} \cdot \boxed{\frac{R_3}{R_1 + R_3}}$

PS5 $\quad \frac{v}{R_1} = \left(V_2 - \frac{V_2}{R_2 + R_1 \| R_3} \cdot R_2\right) \frac{1}{R_1} = \frac{v}{}$

Other

$$I_1 = \frac{V_1 - v}{R_1}$$

$$I_2 = \frac{V_2 - v}{R_2}$$

$$v = R_3\left(\frac{V_1 - v}{R_1} + \frac{V_2 - v}{R_2}\right)$$

$$I_1 = \frac{V_1 - R_3 (I_1 + I_2)}{R_1}$$

$$I_2 = \frac{V_2 - R_3 (I_1 + I_2)}{R_2}$$

PS6 If 2-port passive (R/L/C) : $\underline{y_{12} = y_{21}}$ (Reciprocity theorem)
(No dep source)
I&D

* 3rd prop of Recip ⇒ (Z)

$$\begin{bmatrix} z_{11} & z_{12} & \cdots & \\ & & & z_{ji} \\ & z_{ij} & & \end{bmatrix}$$

$z_{ij} = z_{ji}$ if $\underline{\text{passive}}$

$\neq$ if $\underline{\text{dep source}}$

$y_{12} = \left[Z^{-1}\right]_{12} \quad y_{21} = \left[Z^{-1}\right]_{21}$